An Overview of Interpolation Techniques

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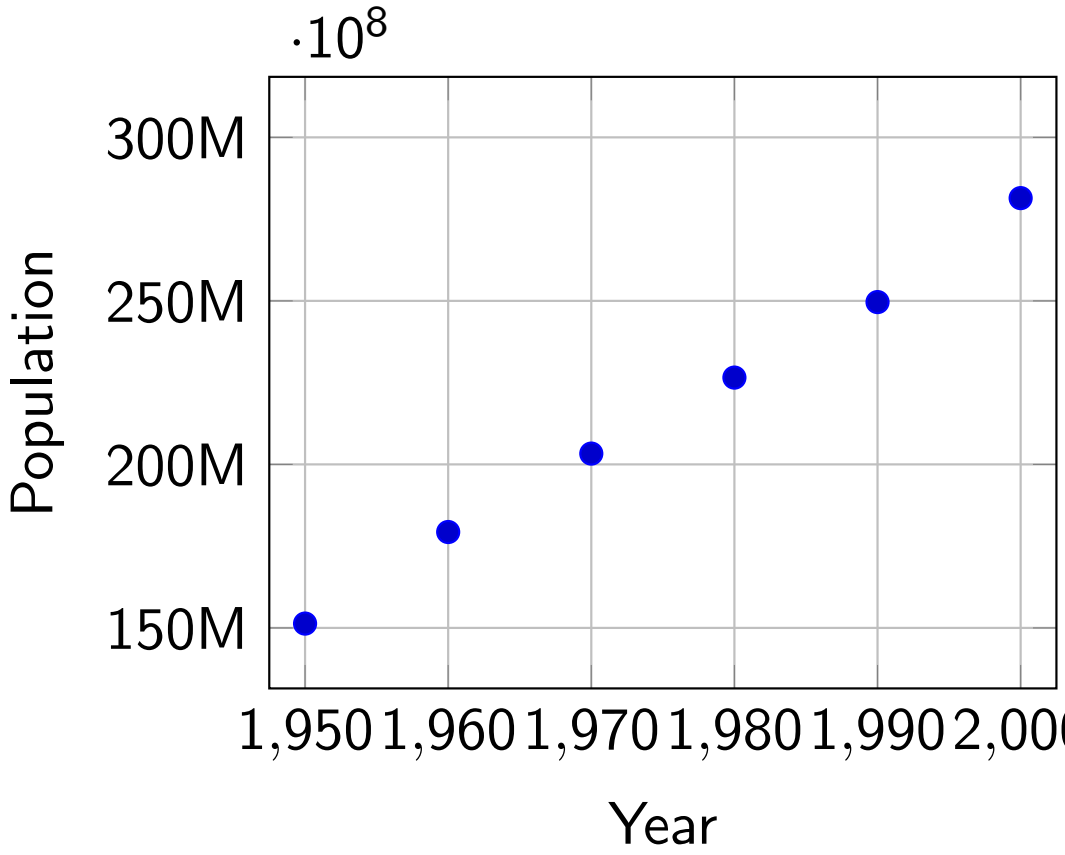
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U.S. Population (1950–2000)

Census Data

Year	Population (thousands)
1950	151,326
1960	179,323
1970	203,302
1980	226,542
1990	249,633
2000	281,422



Context

A census of the U.S. population is conducted every 10 years. This data—from 1950 to 2000—forms the basis for exploring interpolation: for instance, estimating the population in 1975 or projecting it to 2020.

Lagrange Interpolation

Interpolation Setup Given $n+1$ distinct nodes (x_i, y_i) with $y_i = f(x_i)$, we seek a polynomial $p(x) \in \mathcal{P}_n$ such that:

$$p(x_i) = y_i, \quad i = 0, 1, \ldots, n.$$

Lagrange Formula

$$p(x) = \sum_{i=0}^{n} \left(y_i \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} \right).$$

Theorem (Existence & Uniqueness)

There is exactly one such polynomial $p(x)$ of degree at most n.

Lagrange Interpolation — Example (3 points)

Data: $(0,1),\ (1,3),\ (2,2)$.

Lagrange basis:

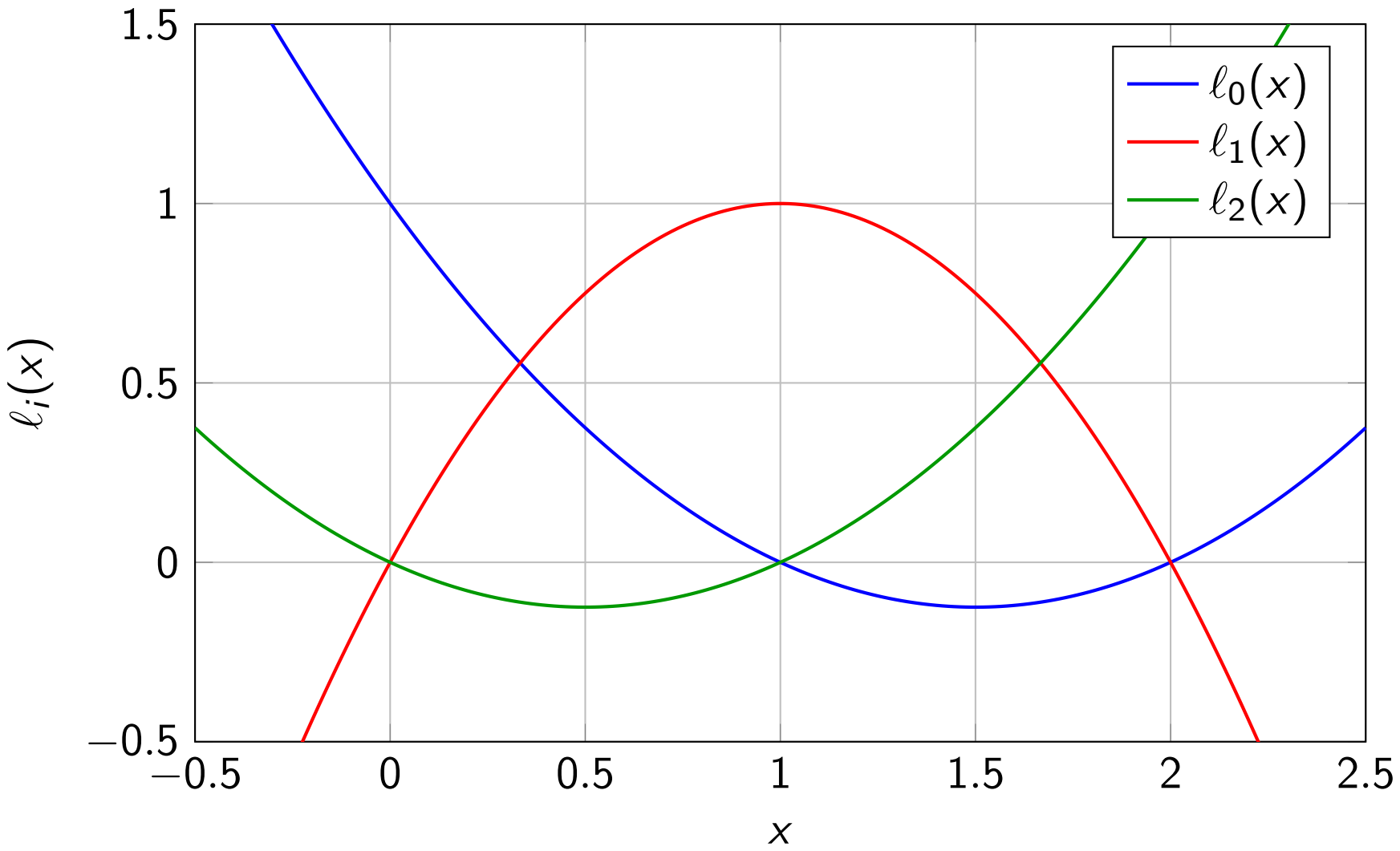
$$\ell_0(x) = \frac{(x-1)(x-2)}{2}, \quad \ell_1(x) = -x(x-2), \quad \ell_2(x) = \frac{x(x-1)}{2}.$$

Interpolant:

$$\begin{aligned} p(x) &= 1 \cdot \ell_0(x) + 3 \cdot \ell_1(x) + 2 \cdot \ell_2(x) \\ &= \frac{(x-1)(x-2)}{2} - 3x(x-2) + \frac{2x(x-1)}{2} \\ &= -\tfrac{3}{2}x^2 + \tfrac{7}{2}x + 1. \end{aligned}$$

Example evaluation: $p(1.5) = 2.875$

Lagrange Basis $\ell_i(x)$ for $(0,1),(1,3),(2,2)$



Observation: Each $\ell_i(x)$ is 1 at x_i and 0 at other nodes, ensuring $\sum_i y_i \ell_i(x_j) = y_j$, hence exact interpolation.

Adding a New Point — The Issue with Lagrange

Scenario. You have a Lagrange interpolant with $n+1$ points:

$$P_n(x) = \sum_{i=0}^{n} y_i \ell_i(x), \quad \ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}.$$

The Problem. - When a new point (x_{n+1}, y_{n+1}) arrives: Every existing basis polynomial $\ell_i(x)$ must be recomputed to accommodate the new node. - That means a complete recalculation - regardless of how many points you already have. - This is inefficient both computationally and numerically.

Conclusion. Lagrange's form doesn't support efficient incremental updates—adding new data forces full recomputation of the interpolant.

Newton's Divided-Difference Form (Reformatted)

The Newton interpolation polynomial can be written as:

$$\begin{aligned} P_n(x) = f[x_0] \\ &+ f[x_0, x_1](x - x_0) \\ &+ f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ &+ \dots \\ &+ f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1}). \end{aligned}$$

Why the Newton Form Is Efficient for Adding Points

Given $P_n(x)$ built from $(x_0, \ldots, x_n)$, adding a new point (x_{n+1}, y_{n+1}) only requires:

$$P_{n+1}(x) = P_n(x) + f[x_0, x_1, \ldots, x_{n+1}] \cdot (x - x_0)(x - x_1) \cdots (x - x_n).$$

- No need to recompute earlier coefficients -just compute one new divided difference and append a term. - Efficient and flexible, ideal when data grows dynamically.
In contrast, the Lagrange form requires re-evaluating all basis polynomials when a new point is added.

Divided Difference Table (3 points)

x	$f[x]$	$f[\,x,\,x\,]$	$f[\,x,\,x,\,x\,]$
0	1	2	-3/2
1	3	-1	
2	2		

Interpretation: the first column is the function values; the first divided differences (col. 2) are slopes between successive points; the leading elements of each column are the coefficients used in the Newton form.

Newton's Polynomial and Checks

Using the table:

$$P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$P_2(x) = 1 + 2(x - 0) - \tfrac{3}{2}(x - 0)(x - 1).$$

Expand:

$$P_2(x) = -\tfrac{3}{2}x^2 + \tfrac{7}{2}x + 1,$$

which matches the Lagrange result.
Numeric evaluation: $P_2(1.5) = 2.875$.

Note: to add a new point (x_3, y_3) you only need the new divided difference $f[x_0, x_1, x_2, x_3]$ and then append the single term

$$f[x_0, \ldots, x_3](x - x_0)(x - x_1)(x - x_2),$$

so previous coefficients remain unchanged (efficient incremental update).

Add a 4th Point — Divided Difference Table

x	$f[x]$	$f[x,x]$	$f[x,x,x]$
0	1	2	-3/2
1	3	-1	2
2	2	3	
3	5		

Only the new column entries (the rightmost values) need computing when we add $(x_3, y_3) = (3, 5)$. The new top coefficient (third divided difference) is $f[x_0, x_1, x_2, x_3] = \frac{7}{6}$.

Updated Newton Polynomial & Checks

Newton form (with the new coefficient):

$$P_3(x) = 1 + 2(x-0) - \tfrac{3}{2}(x-0)(x-1) + \tfrac{7}{6}(x-0)(x-1)(x-2).$$

Expanded:

$$P_3(x) = \tfrac{7}{6}x^3 - 5x^2 + \tfrac{35}{6}x + 1.$$

Checks:

- $P_3(3) = 5$ (recovers the new data point).
- $P_3(1.5) = 39/16 \approx 2.4375$ (note the interpolant changes from the previous quadratic).

$$f[x_0, \ldots, x_4](x - x_0)(x - x_1)(x - x_2)(x - x_3),$$

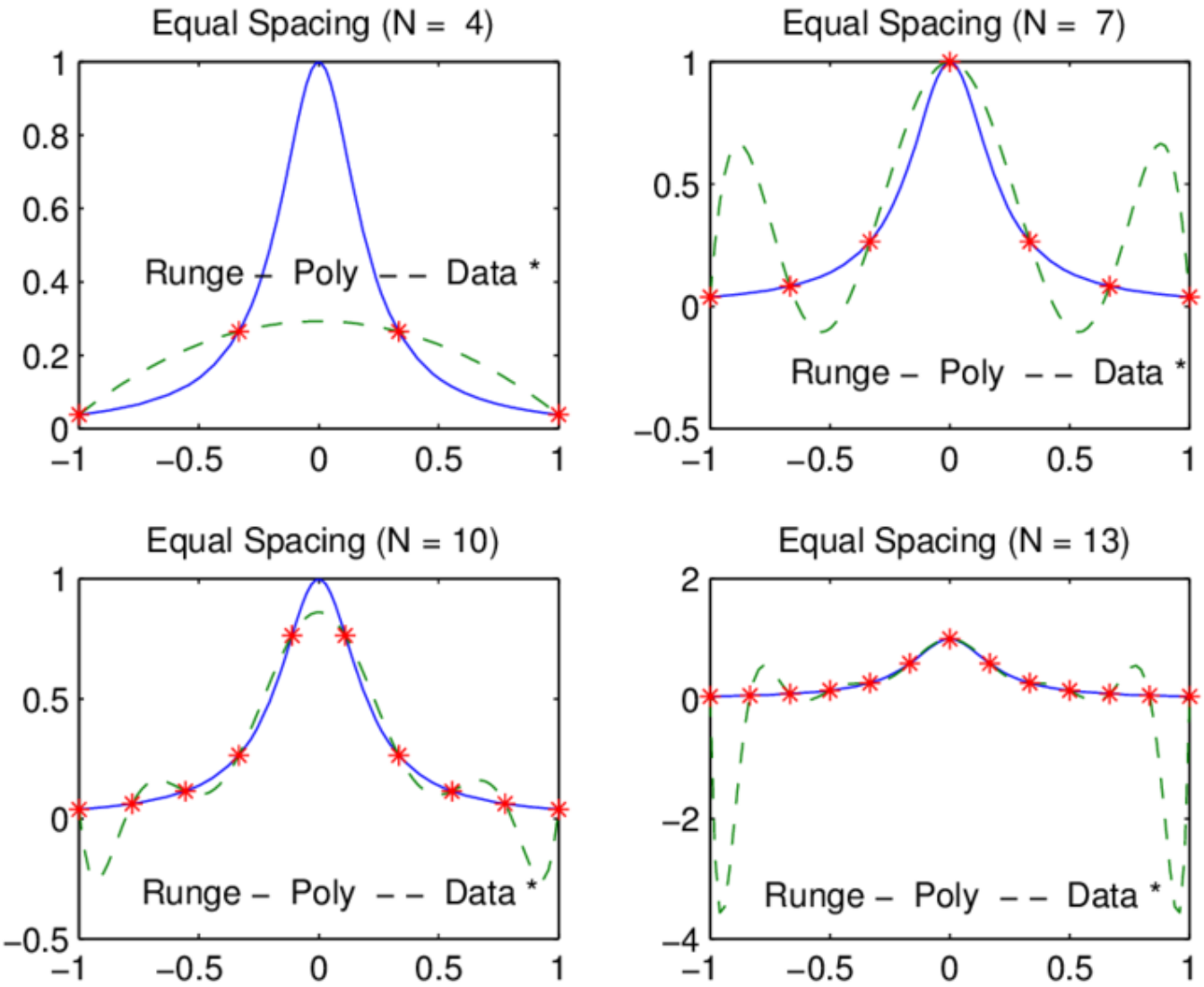
Global Polynomial Interpolation: Challenges

Key Issue: Fitting a single high-degree polynomial to many data points across an interval can lead to serious oscillations near the ends-known as "Runge's phenomenon".
Consider Runge's function:

$$f(x) = \frac{1}{1 + 25x^2}, \quad x \in [-1, 1].$$

Interpolating f at many *equispaced* nodes, one would expect higher-degree polynomials to better approximate f. However, in practice, oscillations near the boundaries grow with increasing polynomial degree.

Runge's Phenomenon in Action



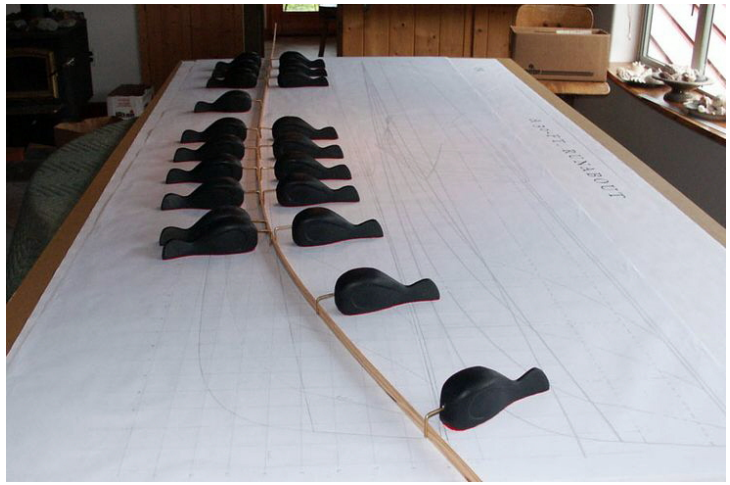
Mitigating Runge's Phenomenon

Here are two reliable ways to avoid oscillatory divergence:

- **Chebyshev Nodes:** Use non-uniform nodes clustered near the boundaries. This minimizes the interpolation error and mitigates oscillations.
- **Piecewise (Spline) Interpolation:** Fit low-degree polynomials locally (e.g. cubic splines). These avoid global oscillation while offering smooth approximations.

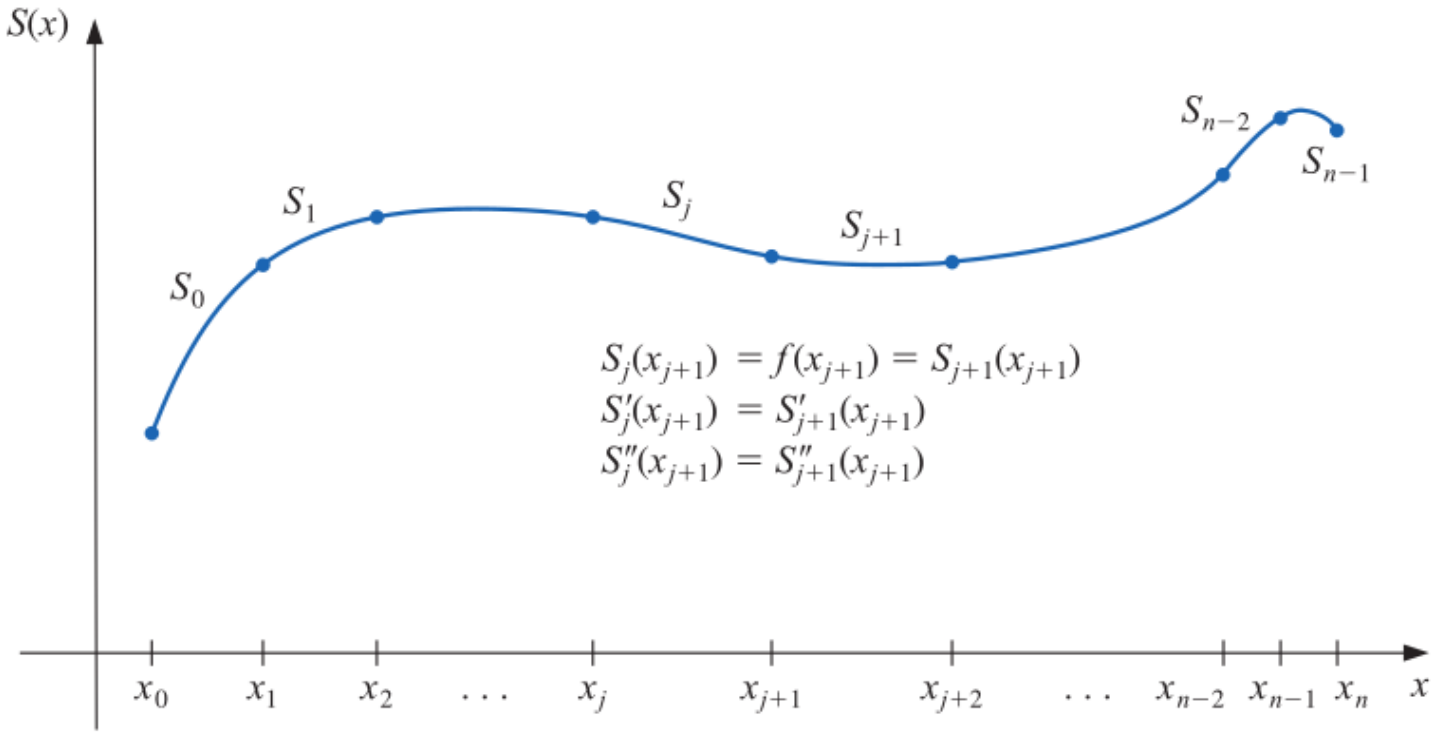
What Is a Spline? History & Origins

Historically, a spline was a flexible wooden or metal strip used by naval architects and aircraft designers to draw smooth curves through designated points—known as lofting or even lofting from shipbuilding. As early as the 18th century, flexible strips were bent against constraints (called "ducks" or "weights") to define smooth shapes. Over time, I. J. Schoenberg (circa 1946) formalized the concept in mathematics, introducing mathematical splines as piecewise polynomial approximations. Since then, splines have become central in CAD, numerical analysis, computer graphics, and curve design.





Cubic Spline Interpolation—Visual Example



- The curve is made of piecewise cubic polynomials (one per interval between data points). - These are joined with C^2 continuity—smooth in value, slope, and curvature across knots. - This avoids the oscillatory issues of high-degree global polynomials.

Definition of Cubic Spline Interpolant

Given $n+1$ data points (x_i, y_i), a "cubic spline" $S(x)$ satisfies:

- On each interval $[x_i, x_{i+1}]$, S is a cubic polynomial S_i.
- Interpolation: $S(x_i) = y_i$, for all i.
- Smoothness: S, S', and S'' are continuous at each internal knot.

Two common boundary conditions:

Natural spline: $S''(x_0) = 0, \quad S''(x_n) = 0.$

Clamped spline: $S'(x_0) = f'(x_0), \quad S'(x_n) = f'(x_n).$

Natural Cubic Spline Example (Setup)

Construct a natural cubic spline through the points $(1, 2)$, $(2, 3)$, and $(3, 5)$.

Step 1: Define cubic polynomials for each interval

$$S_1(x) = a_1 + b_1(x-1) + c_1(x-1)^2 + d_1(x-1)^3, \quad 1 \le x \le 2$$
$$S_2(x) = a_2 + b_2(x-2) + c_2(x-2)^2 + d_2(x-2)^3, \quad 2 \le x \le 3$$

Step 2: Interpolation conditions

$$S_1(1) = 2, \quad S_1(2) = 3$$
$$S_2(2) = 3, \quad S_2(3) = 5$$

Step 3: Continuity conditions at $x = 2$

$$S_1'(2) = S_2'(2), \quad S_1''(2) = S_2''(2)$$

Step 4: Natural spline boundary conditions

$$S_1''(1) = 0, \quad S_2''(3) = 0$$

Natural Cubic Spline Example (Solution)

Step 5: Solve for coefficients

From the above conditions, we find:

$$S_1(x) = 2 + (x - 1) = x + 1$$
$$S_2(x) = 3 + 2(x - 2) - (x - 2)^2$$

Step 6: Final piecewise spline

$$S(x) = \begin{cases} x + 1, & 1 \leq x \leq 2 \\ 3 + 2(x - 2) - (x - 2)^2, & 2 \leq x \leq 3 \end{cases}$$

Thank You!

Questions & Discussion